

Question 1**1.**

- a. The slope coefficient on the excess return of AAPL over MSFT denotes the optimal weight to be allocated to MSFT. The investor ought to allocate 78.41% of their wealth to MSFT and the remaining $100\% - 78.41\% = 21.59\%$ of their wealth to AAPL.

b. We have:

$$\begin{aligned}
 w_{AAPL} &= \frac{\sigma_{MSFT}^2 - \sigma_{MSFT,AAPL}}{\sigma_{AAPL}^2 + \sigma_{MSFT}^2 - 2\sigma_{MSFT,AAPL}} \\
 &= \frac{0.0073 - 0.0041}{0.0158 + 0.0073 - 2 \times 0.0041} \\
 &= 0.2159
 \end{aligned}$$

and

$$\begin{aligned}
 w_{MSFT} &= 1 - w_{AAPL} \\
 &= 1 - 0.2159 = 0.7841.
 \end{aligned}$$

2.

a. We have:

$$r_t - r_{ft} = \beta_1 (r_{mt} - r_{ft}) + \eta_t$$

since $\beta_0 = \beta_2 = \beta_3 = \beta_4 = 0$. Because r_{ft} is non-stochastic (as r_{ft} is time invariant) and $cov(r_{mt} - r_{ft}, \eta_t) = 0$,

$$var(r_t) = \beta_1^2 var(r_{mt}) + var(\eta_t).$$

This shows that the variance of the return on a stock r_t consists of two parts: a part related to the variance of the market index and an idiosyncratic part.

- b. The fraction of total risk accounting for systematic (non-diversifiable risk) can be deduced by the R^2 of regression. We have that $R^2 = 0.6951$ so that 69.51% of total variation in $(r_t - r_{ft})$ corresponds to systematic risk, while the remaining $1 - R^2 \approx 0.3049$ (30.49%) of total return variation is portfolio specific.

3.

- a. The coefficient estimate attached to the size (SMB) factor stands at $0.549 > 0$ suggesting that the durables portfolio behaves more like a small-stock portfolio. To check the significance of this factor in explaining the variation in excess returns of this portfolio, we test the following hypothesis:

$$H_0 : \beta_2 = 0 \text{ vs } \beta_2 \neq 0$$

and use the t-statistic:

$$t\text{-stat} = \frac{\hat{\beta}_2 - 0}{se(\hat{\beta}_2)} = \frac{0.54912}{0.15918} = 3.45.$$

This statistic follows a t-distribution under the null. Comparing with the (2-tailed) critical value from the t-distribution at 1% significance level (and ∞ degrees of freedom (218-4-1=213 degrees of freedom) we have that $|t\text{-stat}| > 2.576$, and we conclude that we reject the null. We conclude that the SMB factor makes a significant contribution in explaining the dependent variable.

- b. In order to test for joint significance for the four factors over that with one (the market factor), we define the following null and alternative hypotheses:

$$\begin{aligned} H_0 &: \beta_2 = \beta_3 = \beta_4 = 0 \\ H_1 &: \text{at least one is not zero} \end{aligned}$$

In the R output,

$$\begin{aligned} \text{Residual standard error}_1 &= \sqrt{\frac{RSS_1}{T - k - 1}} \\ 5.33 &= \sqrt{\frac{RSS_1}{218 - 4 - 1}} \\ RSS_1 &= 6051.66 \end{aligned}$$

and

$$\begin{aligned} \text{Residual standard error}_0 &= \sqrt{\frac{RSS_0}{T - k - 1}} \\ 5.728 &= \sqrt{\frac{RSS_0}{218 - 1 - 1}} \\ RSS_0 &= 7087.26 \end{aligned}$$

and use the J-statistic:

$$\begin{aligned} J &= \frac{RSS_0 - RSS_1}{RSS_1 / (T - k - 1)} \\ &= \frac{7087.26 - 6051.660}{6051.660 / (218 - 4 - 1)} = 36.45 \end{aligned}$$

where RSS_0 and RSS_1 denote the residual sum of squares from running the constrained version of (1) (CAPM - three restrictions imposed), and (1) without restrictions, respectively. T denotes the number of observations, and k are the number of regressors in the unrestricted model. This statistic follows a chi-squared distribution with 3 degrees of freedom under the null. We compare the calculated value for J with the relevant critical value from the χ^2_3 -distribution at 5% significance level ($= 7.81$). J -calculated is larger than the chi-squared critical value so we reject the null, i.e. the 4FF specification explains more adequately than the CAPM model the systematic risk inherent in the durables portfolio excess returns.

Question 2

1. In order to test for existence of a unit root in p_t we conduct an Augmented Dickey Fuller test which uses the following regression:

$$\Delta p_t = \omega_0 + \omega_1 t + \phi_{c1} p_{t-1} + \sum_{i=1}^{p-1} \beta_i \Delta p_{t-i} + \varepsilon_t,$$

We have the following null and alternative hypotheses:

$$\begin{aligned} H_0 &: \phi_{c1} = 0 \\ H_1 &: \phi_{c1} < 0 \end{aligned}$$

The ADF t-ratio is then given by:

$$ADF \equiv T = \frac{\hat{\phi}_{c1}}{se(\hat{\phi}_{c1})}.$$

Given the results in Figure 3(a), we have:

$$ADF \equiv T = \frac{\hat{\phi}_{c1}}{se(\hat{\phi}_{c1})} = -2.402$$

The 5% critical value is $\tau_3 = -3.41$. Since $ADF = -2.402 > -3.41$ the null hypothesis of existence of a unit root cannot be rejected. The process p_t appears to be non-stationary.

2. Since u_t is an MDS, $E(u_t|F_{t-1}) = 0$.

(a) The conditional mean

$$\begin{aligned} E(r_t|F_{t-1}) &= E(0.1 + 0.5r_{t-1} + u_t|F_{t-1}) \\ &= E(0.1|F_{t-1}) + E(0.5r_{t-1}|F_{t-1}) + E(u_t|F_{t-1}) \\ &= 0.1 + 0.5r_{t-1} \end{aligned}$$

(b) The unconditional mean (using LIE)

$$\begin{aligned} E(r_t) &= E(E(r_t|F_{t-1})) \\ &= E(0.1 + 0.5r_{t-1}) \\ &= 0.1 + 0.5E(r_{t-1}) \\ &= 0.1 + 0.5E(r_t) \quad r_t \text{ is stationary} \\ E(r_t) &= \frac{0.1}{1 - 0.5} = 0.2 \end{aligned}$$

(c) The conditional variance

$$\begin{aligned} \mathbb{V}(r_t|F_{t-1}) &= \mathbb{V}(0.1 + 0.5r_{t-1} + u_t|F_{t-1}) \\ &= \mathbb{V}(u_t|F_{t-1}) \\ &= 4 \end{aligned}$$

since r_{t-1} is known given F_{t-1} .

(d) The unconditional variance

$$\begin{aligned}
 \mathbb{V}(r_t) &= E(\mathbb{V}(r_t|F_{t-1})) + \mathbb{V}(E(r_t|F_{t-1})) \\
 &= E(4) + \mathbb{V}(0.1 + 0.5r_{t-1}) \\
 &= 4 + 0.5^2 \mathbb{V}(r_{t-1}) \\
 &= 4 + 0.5^2 \mathbb{V}(r_t) \quad r_t \text{ is stationary} \\
 \mathbb{V}(r_t) &= \frac{4}{1 - 0.5^2} = 5.333
 \end{aligned}$$

(e) Yes, because using F_{t-1} reduces the uncertainty about r_t , so $\mathbb{V}(r_t|F_{t-1}) < \mathbb{V}(r_t)$.

3. The fitted model is an AR(1)-GARCH(1,1) model

$$\begin{aligned}
 r_t &= 0.0899 - 0.0788(r_{t-1} - 0.0899) + \hat{u}_t \\
 \hat{\sigma}_t^2 &= 0.1634 + 0.1038\hat{u}_{t-1}^2 + 0.8488\hat{\sigma}_{t-1}^2
 \end{aligned}$$

(Week 9) Alternatively the mean equation can be written in the usual intercept form

$$r_t = \hat{c} - 0.0788r_{t-1} + \hat{u}_t$$

where

$$\hat{\mu} = \frac{\hat{c}}{1 - \hat{\phi}_1}$$

and so

$$\hat{c} = \hat{\mu}(1 - \hat{\phi}_1) = 0.0899(1 + 0.0788) = 0.0970$$

and so

$$r_t = 0.0970 - 0.0788r_{t-1} + \hat{u}_t$$

The 1-step ahead forecasts are

$$\begin{aligned}
 E(r_{t+1}|F_t) &= 0.0899 - 0.0788(r_t - 0.0899) \\
 &= 0.0899 - 0.0788(0.309 - 0.0899) \\
 &= 0.0727\%
 \end{aligned}$$

Alternatively using $r_t = 0.0970 - 0.0788r_{t-1} + \hat{u}_t$, the 1-step ahead forecasts are

$$\begin{aligned}
 E(r_{t+1}|F_t) &= 0.0970 - 0.0788r_T \\
 &= 0.0970 - 0.0788(0.309) = 0.0727\%
 \end{aligned}$$

And

$$\begin{aligned}
 E(\hat{\sigma}_{t+1}^2|F_t) &= 0.1634 + 0.1038\hat{u}_t^2 + 0.8488\hat{\sigma}_t^2 \\
 &= 0.1634 + 0.1038(0.2660^2) + 0.8488(4.6638) \\
 &= 4.1296
 \end{aligned}$$

The 95% prediction interval of r_t is

$$0.0727 \pm 1.96\sqrt{4.1296}$$

giving the interval

$$[-3.9104\%, 4.0557\%]$$

Note: Minor variations in answers may occur due to rounding.

4. The 1-day-ahead 5% VaR is

$$\begin{aligned}q_{0.05} &= 0.0727 + \sqrt{4.1296}(-1.645) \\&= -3.2701\% \quad [\text{note: \%}]\end{aligned}$$

and so

$$\begin{aligned}\text{VaR}_{0.05} &= |1,000,000(-0.032701)| \\&= \$32,701\end{aligned}$$

Note: Minor variations in answers may occur due to rounding.

Question 3

1.

a. After running the constant mean equation:

$$r_{V,t} = c + \nu_t,$$

we consider the following auxiliary regression:

$$\hat{\nu}_t^2 = \gamma_0 + \gamma_1 \hat{\nu}_{t-1}^2 + \gamma_2 \hat{\nu}_{t-2}^2 + \dots + \gamma_5 \hat{\nu}_{t-5}^2 + \eta_t.$$

We test the null hypothesis:

$$H_0 : \gamma_1 = \gamma_2 = \dots = \gamma_5 = 0 \text{ [No ARCH]}$$

vs

$$H_1 : \text{At least one differs from zero [ARCH]}$$

The test statistic is given by:

$$ARCH(5) = TR^2 = 1135.6$$

where R^2 is the fit of the auxiliary regression and T is the sample size. This follows a χ_5^2 distribution asymptotically. From the results of Figure 4 there is evidence that the null of no ARCH effects is rejected at 1%, 5% and 10% significance level (when comparing to the respective critical values and/or using the p-values). This implies that the investor can improve her modelling strategy which includes a conditional variance equation for this process using an ARCH model or generalisation thereof.

b. Unlike the ARCH test, the White procedure tests for heteroskedasticity that is not linked to past information (F_{t-1}) but is related to the explanatory variables included in the model.

2.

a. We have:

$$\begin{aligned} r_{V,t} &= c + \nu_t, \\ \nu_t &= \sigma_t \varepsilon_t, \quad \varepsilon_t \sim i.i.d.N(0, 1) \\ \sigma_t^2 &= a_0 + a_1 \nu_{t-1}^2 + \beta_1 \sigma_{t-1}^2 \end{aligned}$$

The model requires that $a_0, a_1, \beta_1 > 0$ to ensure that $\sigma_t^2 > 0$, and $a_1 + \beta_1 < 1$, to ensure that the unconditional variance $V(\nu_t)$ exists. From the results of Figure 5 we have that $\hat{a}_0 = 0.000009$, $\hat{a}_1 = 0.08078663$, $\hat{\beta}_1 = 0.9040081$, and $\hat{a}_1 + \hat{\beta}_1 = 0.9848 < 1$; ie. the estimates comply with the conditions of the model.

b. From the LIE we have that:

$$E(\nu_t^3) = E[E(\nu_t^3 | F_{t-1})]$$

and since $\nu_t | F_{t-1} \sim N(0, \sigma_t^2)$ and hence $E(\nu_t^3 | F_{t-1}) = 0$. Then,

$$E(\nu_t^3) = E[E(\nu_t^3 | F_{t-1})] = E[0] = 0,$$

hence the unconditional distribution is symmetric. (Week 2) The unconditional skewness of v_t is

$$\frac{E[\nu_t - E(\nu_t)]^3}{V(\nu_t)^{3/2}} = \frac{E[\nu_t^3]}{V(\nu_t)^{3/2}} = 0$$

because $E[\nu_t^3] = 0$ and $E(v_t) = E(E(v_t | F_{t-1})) = 0$.

3.

a. We have:

$$\begin{aligned} r_{V,t} &= c + \nu_t, \\ \nu_t &= \sigma_t \varepsilon_t, \quad \varepsilon_t \sim i.i.d.N(0,1) \\ \sigma_t^2 &= a_0 + a_1 \nu_{t-1}^2 + \lambda (I_{t-1} \nu_{t-1}^2) + \beta_1 \sigma_{t-1}^2 \end{aligned}$$

and

$$I_{t-1} = \begin{cases} 1, & \text{if } \nu_{t-1} \leq 0 \\ 0, & \text{otherwise} \end{cases}.$$

Compared with the GARCH(1,1) model, $\lambda > 0$ implies that negative news have a larger impact on volatility than positive news, indicating a leverage/asymmetric volatility effect.

b. The NIC for the GJR-GARCH(1,1,1) model can be written as:

$$\sigma_t^2 = \begin{cases} A + a_1 \nu_{t-1}^2, & \text{if } \nu_{t-1} > 0 \\ A + (a_1 + \lambda) \nu_{t-1}^2, & \text{if } \nu_{t-1} \leq 0 \end{cases}$$

where

$$A = a_0 + \beta_1 \frac{a_0}{1 - (a_1 + \lambda/2) - \beta_1}.$$

With $a_1, \lambda \geq 0$ and $a_1 + \lambda \geq 0$, it is clear that the curvature of the NIC changes depending on whether we have a positive or negative shock materialising. More precisely, it will be steeper $(a_1 + \lambda)$ when $\nu_{t-1} < 0$ and flatter a_1 when $\nu_{t-1} > 0$. This is confirmed by the numerical values of these parameters from the results of Figure 6. The NIC slope amounts to $(\hat{a}_1 + \hat{\lambda}) = (0.05486562 + 0.05483184) = 0.1097$ when $\nu_{t-1} < 0$ and $\hat{a}_1 = 0.05486562$ when $\nu_{t-1} > 0$. Since $\hat{\lambda} > 0$ the news impact curve is steeper for negative shocks than for positive shocks. This implies that negative shocks increase conditional volatility more than positive shocks of the same magnitude.

Question 4

1. We run the following auxiliary regression:

$$\hat{u}_t = \gamma_0 + \gamma_1(r_{mt} - r_{ft}) + \gamma_2\text{SMB}_t + \gamma_3\text{HML}_t + \gamma_4\text{MOM}_t + \rho_1\hat{u}_{t-1} + \rho_2\hat{u}_{t-2} + \dots + \rho_p\hat{u}_{t-p} + \varepsilon_t,$$

or

$$\hat{u}_t = \gamma_0 + \rho_1\hat{u}_{t-1} + \rho_2\hat{u}_{t-2} + \dots + \rho_p\hat{u}_{t-p} + \varepsilon_t,$$

where lag length $p = 2$. We then test the following null hypothesis:

$$H_0 : \rho_1 = \rho_2 = \dots = \rho_p = 0$$

versus

$$H_1 : \text{at least one is not zero}$$

The relevant statistic is $AR(p) = TR^2$ where R^2 is the R-squared of the auxiliary regression. This follows a χ^2 distribution with p degrees of freedom under the null. If the null hypothesis is rejected then this is indication that the EMH does not hold. Otherwise, we have evidence that the EMH holds over this sample period.

2. From the results of Figure 7, p is set to 2 and $AR(2) = TR^2 = 64.641$. Comparing this with the χ^2 distribution with 2 degrees of freedom at 5% significance level ($=5.99$), it is clear that $AR(2) > \chi^2_2$ and the null hypothesis is rejected. Therefore, the EMH does not hold over this sample period.
3. There are three versions of the EMH. They differ from each other in terms of the information set on which they are defined. We have: (i) the weak form: prices reflect the information in all past prices; (ii) semi-strong form: prices reflect all publicly available information; (iii) strong form: prices reflect all acquirable information.

Question 5

1. Denoting the Nikkei log returns by r_t , the model is written as:

$$\begin{aligned}r_t &= \phi_0 + \varepsilon_t + \theta_1 \varepsilon_{t-1} \\ \varepsilon_t &= \sigma_t u_t, \quad u_t \sim i.i.d.N(0, 1) \\ \sigma_t^2 &= a_0 + a_1 \varepsilon_{t-1}^2 + a_2 \varepsilon_{t-2}^2 + a_3 \varepsilon_{t-3}^2 + a_4 \varepsilon_{t-4}^2 + a_5 \varepsilon_{t-5}^2,\end{aligned}$$

and hence

$$\begin{aligned}r_t &= 0.0531 + \hat{\varepsilon}_t - 0.0224 \hat{\varepsilon}_{t-1} \\ \hat{\sigma}_t^2 &= 0.7295 + 0.1074 \hat{\varepsilon}_{t-1}^2 + 0.1992 \hat{\varepsilon}_{t-2}^2 + 0.1671 \hat{\varepsilon}_{t-3}^2 + 0.1111 \hat{\varepsilon}_{t-4}^2 + 0.1160 \hat{\varepsilon}_{t-5}^2\end{aligned}$$

2. The model is written as:

$$\begin{aligned}r_t &= c + \varepsilon_t, \\ \varepsilon_t &= \sigma_t u_t, \quad u_t \sim i.i.d.N(0, 1) \\ \sigma_t^2 &= a_0 + a_1 \varepsilon_{t-1}^2 + a_2 \varepsilon_{t-2}^2 + a_3 \varepsilon_{t-3}^2 + a_4 \varepsilon_{t-4}^2 + a_5 \varepsilon_{t-5}^2,\end{aligned}$$

and hence

$$\begin{aligned}r_t &= 0.0535 + \hat{\varepsilon}_t \\ \hat{\sigma}_t^2 &= 0.7310 + 0.1073 \hat{\varepsilon}_{t-1}^2 + 0.1981 \hat{\varepsilon}_{t-2}^2 + 0.1685 \hat{\varepsilon}_{t-3}^2 + 0.1109 \hat{\varepsilon}_{t-4}^2 + 0.1150 \hat{\varepsilon}_{t-5}^2\end{aligned}$$

3. We notice that models GARCH and ARCH are non-nested since there is no set of restrictions on the parameters of the GARCH model that can be imposed to obtain the ARCH model. For this reason comparison between the performance of the two models needs to be determined by use of the Akaike Information Criterion (AIC). Comparing the AIC from Figures 9 and 10 we see that ARCH has a value of 3.429337 and GARCH has a value of 3.386934. The GARCH model outperforms since it has the lowest AIC reading.